PRCP-1000 Portuguese Bank Marketing

Domain: Finance / Banking

Problem Statement

A Portuguese bank did direct phone call marketing campaigns from May 2008 to November 2010 to promote term deposits among its existing customers. Calling every customer takes a lot of time and money, and most of the customers do not subscribe. The bank marketing team wants to know which customers are likely to buy the product (subscribe to a term deposit) so that they can call the right customers.

Tasks

Task 1: Prepare a complete data analysis report on the given data.

Task 2: Create a predictive model which will help the bank marketing team to know which customer will buy the product.

Task 3: Give suggestions to the bank marketing team to make customers buy the product.

Dataset

The dataset is about the direct phone call marketing campaigns of the bank. The file **bank-additional-full.csv** (inside Data > bank-additional) is used for this project. It has 41,188 rows and 21 columns, and the values are separated by semicolons.

Link: https://d3ilbtxij3aepc.cloudfront.net/projects/CDS-Capstone-Projects/PRCP-1000-ProtugeseBank.zip

Target variable: **y** - has the client subscribed to a term deposit? (yes / no)

Attribute Information

Client details:

- age - age of the client (numeric)
- job - type of job (admin., blue-collar, entrepreneur, housemaid, management, retired, self-employed, services, student, technician, unemployed, unknown)
- marital - marital status (divorced, married, single, unknown)
- education - education level (basic.4y, basic.6y, basic.9y, high.school, illiterate, professional.course, university.degree, unknown)
- default - has credit in default? (no, yes, unknown)
- housing - has housing loan? (no, yes, unknown)
- loan - has personal loan? (no, yes, unknown)

Last contact of the current campaign:

- contact - contact communication type (cellular, telephone)
- month - last contact month of the year
- day_of_week - last contact day of the week

- duration - last contact duration in seconds. This is not known before a call is made, so it should not be used for a realistic model.

Other attributes:

- campaign - number of contacts made during this campaign for this client
- pdays - number of days since the client was last contacted in a previous campaign (999 means the client was not contacted before)
- previous - number of contacts made before this campaign
- poutcome - outcome of the previous marketing campaign (failure, nonexistent, success)

Social and economic attributes:

- emp.var.rate - employment variation rate (quarterly)
- cons.price.idx - consumer price index (monthly)
- cons.conf.idx - consumer confidence index (monthly)
- euribor3m - euribor 3 month rate (daily)
- nr.employed - number of employees (quarterly)

Model Comparison Report

Create a report stating the performance of multiple models on this data and suggest the best model for production.

Report on Challenges Faced

Create a report which includes the challenges faced on the data and the technique used for each, with proper reason.

Note

All the above tasks are done in a single Jupyter notebook and shared as the final submission of the project.